LEONARDO COLÒ

NUMBER THEORY & CRYPTOGRAPHY

DOCTOR OF PHILOSOPHY IN MATHEMATICS

Date of birth: Oct 14, 1994
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I am currently a Postdoc in the research group of Professors David Jao and Douglas Stebila. My research interests lie at the intersection of number theory and cryptography. In particular, I investigate various aspects of isogeny graphs and their potential cryptographic applications.

WORK EXPERIENCE

2023-present **Postdoctoral Fellow.** University of Waterloo, ON Canada.
- Program: *Accelerating the transition to quantum-resistant cryptography*.

Summer 2026 **Internship.** Qorsa, Waterloo, ON, Canada.
- Building a Go-based TLS scanner for post-quantum migration.

2021-2023 **Teaching and research assistant (ATER).** Aix-Marseille Université.

EDUCATION

2018-2022 **PhD in Mathematics and Cryptography.** Aix-Marseille Université.
- Thesis Title: *Oriented supersingular elliptic curves and class group actions.*

2016-2018 **Master's Degree in Mathematics.** ALGANT Master Program.
- Concordia University (Montréal, CA) & University of Milan (Milano, IT). **GPA: 4.14**

2013-2016 **Bachelor's Degree in Mathematics.** University of Milan.
- Final Seminar: *Intersection of Ideals in a Multivariable Polynomial Ring*. **Grade: 110/110**

2008-2013 **High School Diploma.** Liceo Scientifico Statale "Leonardo da Vinci", Milano (Italy).
- Final research: *Game theory and military strategy*. **Grade: 100/100**

AWARDS

Honors Member of the Concordia University Chapter of the Golden Key International Honour Society.

Scholarships Concordia International Tuition Award of Excellence, Concordia University, 2016/2017.

PUBLICATIONS

complete list of projects available at https://leonardocolo.com/research.html

2026 **Zero-Knowledge proofs of isogeny diamonds.** ePrint
With M. Mamah, Y. Mokrani, B. Sterner, N. Swanson. Communications in Cryptology, vol 3.3

2026 **Weber modular curves and modular isogenies.** ArXiv
With David Kohel. Polynesian Journal of Mathematics, vol 4.13, pp. 1–22

2022 **On a modular approach to the OSIDH protocol.** Preprint
With David Kohel.

2020 **Orienting supersingular isogeny graphs.** arXiv
With David Kohel. Journal of Mathematical Cryptology, vol 14.1, pp. 414–437

INVITED TALKS & VISITS

complete list of talks available at https://leonardocolo.com/talks.html

Nov. 2026 **Research visit.** Université Grenoble Alpes, Grenoble, France.

Oct. 2026 **Isogenies, Cryptography & Explicit Arithmetic Geometry.** Lleida, Spain.

Jul. 2026 **Foundations of Computational Mathematics.** Vienna, Austria.

Jun. 2026 **Research visit.** University of Ottawa, ON, Canada.

Dec. 2025 **Research visit.** Aix-Marseille University, Marseille, France.

2025 & 2027 **Workshop on Isogeny Graphs in Cryptography.** Banff, AL, Canada.

TEACHING EXPERIENCE

complete list of courses available at https://leonardocolo.com/teaching.html

2024–2026 **Course Instructor.** University of Waterloo.

Jan. 2025 **Group Instructor and Supervisor.** CIMPA School, Kampala, Uganda.

2019–2023 **Course Instructor.** Aix-Marseille Université.

2016–2017 **Course Instructor and Teaching Assistant.** Concordia University.

2016 **Teaching Internship.** "Leonardo da Vinci" High school, Milan, Italy.

ACADEMIC ACTIVITIES

2025/2026 Organizer of the reading groups in Cryptography, University of Waterloo.

2024/2025 Organizer of the crypto research group seminar, University of Waterloo.

PROFESSIONAL DEVELOPMENT

2026 **Information Security Specialist Program.** Fields Institute.
- Cryptography, network security, machine learning, quantum computing, risk management.

2026 **CryptoWorks21.** University of Waterloo.
- Cryptography, network security, quantum information, standards and certification, intellectual property protection, and commercialization.

Schools
- Cryptography and coding theory on curves and surfaces (Marseille, France), April 2019
- Mathematical foundations of asymmetric cryptography (Aussois, France), March 2019.

Conferences
- Quantum Consortium Day, February 2025 (University of Toronto, Canada).
- Aix-Marseille Cyber Security Forum, April 2021 & May 2022 (Marseille, FRA).

LANGUAGES

Mother tongue Italian.

Proficient English, French.

PROGRAMMING SKILLS

Programming HTML, CSS, Python, Sage, Magma, Go, LaTeX, AWS, Terraform.

Certifications ECDL Certificate, HTML & CSS CCA Level 1.

OTHER SKILLS

Driving licence **Italy:** A2, B **Canada:** G

Sport **Judo.** International competitions, Italian and Canadian national championships.